



Our Lady of Fatima University

Hilltop Subdivision, Lagro Quezon City

College of Computer Studies

# “An Integrated Barangay Business Permit Application, Review, and Approval Tracking System with Rule-Assisted Scoring Model Using Python”

## **Members:**

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## **Section:**

BSIT 3 - Y2 - 6

## **Instructor:**

Mr. Jordan Cabading



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### Scope

The Integrated Barangay Business Permit Application, Review, and Approval Tracking System is designed to digitalize and streamline the process of applying for and approving business permits at the barangay level. The system allows business owners to submit applications online, upload required documents, and monitor the status of their permit in real time.

Barangay officials can review applications, verify submitted documents, approve or reject requests, and generate reports through a centralized dashboard. The system also includes an approval tracking feature that records processing time, decision history, and responsible personnel for transparency and accountability.

Machine learning, implemented using Python, is integrated to assist in application assessment. The model analyzes historical permit data to help predict application completeness, flag potentially problematic or high-risk applications, and recommend priority levels for review. This feature supports decision-making but does not replace human judgment.

The system covers only barangay-level business permits and is intended for small-scale businesses operating within the barangay's jurisdiction. It is accessible through a web-based interface and requires internet connectivity.

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### **In Scope**

The system will include the following features and functionalities:

#### **Online Business Permit Application**

- Allows business owners to submit permit applications electronically without visiting the barangay office.

#### **User Registration and Authentication**

- Enables secure account creation and login for applicants and barangay officials using role-based access.

#### **Document Upload and Management**

- Allows applicants to upload required documents (e.g., IDs, clearances) and enables officials to view and manage them digitally.

#### **Application Review Module**

- Provides barangay staff with tools to review, verify, and evaluate submitted applications efficiently.

#### **Approval and Rejection Workflow**

- Allows authorized officials to approve or reject applications with recorded remarks and decision history.

#### **Application Status Tracking**

- Enables applicants to monitor real-time updates on the status of their business permit application

#### **Machine Learning–Based Application Assessment**

- Uses Python-based machine learning models to analyze historical data and assist in identifying incomplete or high-risk applications.



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### **Priority Recommendation System**

- Suggests which applications should be reviewed first based on predicted urgency or complexity.

### **Notification System**

- Sends automatic email or system notifications to inform users about application updates, approvals, or rejections.

### **Dashboard and Analytics**

- Displays summaries such as pending applications, approval rates, and processing times for barangay officials.

### **Audit Trail and Activity Logs**

- Records all system actions and transactions to ensure transparency and accountability.

### **Report Generation**

- Generates printable reports on issued permits, pending requests, and historical records.

### **Data Storage and Backup**

- Securely stores application data and ensures backup to prevent data loss.

### **Role-Based Access Control**

- Restricts system features based on user roles (e.g., applicant, staff, barangay captain).

### **System Administration Module**



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- Allows administrators to manage users, system settings, and machine learning model updates.

### Limitations

The system is limited to barangay business permit processing and does not handle permits issued by municipal, city, or national government agencies. Integration with external government systems such as BIR, DTI, or SEC is not included.

The accuracy of the machine learning component depends heavily on the availability and quality of historical data. Incomplete, outdated, or biased data may affect prediction results. The machine learning model provides recommendations only and cannot make final approval decisions.

The system requires stable internet access and basic digital literacy from both applicants and barangay staff. Technical issues such as server downtime, system maintenance, or cybersecurity threats may temporarily affect system availability.

The initial implementation may not fully accommodate all unique barangay rules or frequent policy changes without system updates. Additionally, advanced analytics and large-scale deployment may require higher computational resources.



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January 26, 2026

### **HON. MICHAEL T. PUNONGBAYAN**

Barangay Chairperson

Barangay 183 Amparo, North Caloocan

Dear Sir,

We, the undersigned students of Our Lady of Fatima University – Lagro, Quezon City, respectfully request permission from the Barangay Officials of Barangay 183 Amparo, North Caloocan to conduct our project entitled "An Integrated Barangay Business Permit Application, Review, and Approval Tracking System with Rule-Assisted Scoring Model Using Python"

This project aims to develop a desktop-based system to help the barangay efficiently record, manage, review, monitor, and generate reports for business permit applications in a secure and organized manner. The system is intended for academic purposes only and will be accessible solely to authorized barangay personnel, with all data handled confidentially.

We respectfully seek your approval to propose, develop, and demonstrate this system. Your support will greatly contribute to the success of our project and the improvement of the barangay's record management. We assure that the project will be conducted professionally and following all barangay guidelines.

Thank you for your time and kind consideration. We look forward to your favorable response.

Respectfully yours,

### **ALCANTARA, MIRA JULIANA L.**

Student Researcher

Our Lady of Fatima University

### **CASIMIRO, KYRON MIGUEL C.**

Student Researcher

Our Lady of Fatima University

### **CAWAN, HAZEL JANE A.**

Student Researcher

Our Lady of Fatima University

### **GABUAT, ERICZON M.**

Student Researcher

Our Lady of Fatima University

### **GONZAGA, MHERIEL A.**

Student Researcher

Our Lady of Fatima University

Approved by:

**MR. JORDAN CABADING**



## Our Lady of Fatima University

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College of Computer Studies

January 27, 2026

**HON. MICHAEL T. PUNONGBAYAN**

Barangay Chairperson

Barangay 183 Amparo, North Caloocan

**TO WHOM IT MAY CONCERN:**

I, **HON. MICHAEL T. PUNONGBAYAN**, Barangay Captain of Barangay 183 Amparo, North Caloocan, hereby grant permission to the students of Our Lady of Fatima University – Lagro, Quezon City to conduct their capstone project entitled “An Integrated Barangay Business Permit Application, Review, and Approval Tracking System with Rule-Assisted Scoring Model Using Python” within our barangay.

This approval is given under the condition that the students adhere to all local ordinances, safety protocols, and guidelines set by the barangay. The project may involve the observation, recording, and testing of barangay blotter processes solely for academic purposes. All data collected shall remain confidential and handled responsibly by the students.

We trust that this project will contribute positively to the improvement of our barangay’s record management system.

**Conforme:**

**HON. MICHAEL T. PUNONGBAYAN**

Barangay Chairperson

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DATE